



End-to-End Workflow

The roadmap defines a **structured 16-phase pipeline**:

1. **Business Understanding** → Define churn problem.
2. **Data Understanding** → Load, inspect, classify features.
3. **Data Quality** → Handle missing/duplicate/problematic values.
4. **EDA** → Explore distributions, churn relationships, insights.
5. **Feature/Target Definition** → Explicitly set X (features), y (target).
6. **Train/Test Split** → Before preprocessing.
7. **Preprocessing** → Numerical (imputation, scaling), categorical (encoding), pipeline build.
8. **First Model** → Logistic Regression (train, predict, evaluate).
9. **Evaluation** → Confusion matrix, accuracy, precision, recall, F1, ROC-AUC.
10. **Model Comparison** → Decision Tree, Random Forest, compare metrics.
11. **Model Interpretation** → Feature importance, model behavior.
12. **Final Model Selection** → Document choice, limitations.
13. **Save Model** → Export pipeline (telco_churn_model.pkl).
14. **Application Build** → app.py for predictions with probability + recommended action.
15. **Testing** → Realistic customer scenarios.
16. **Launcher** → .bat file for easy execution.

Final Project Structure:

- Telco_Customer_Churn_Classification.ipynb
- telco_churn_model.pkl
- app.py
- Start_Telco_Churn_Predictor.bat